

The background is a dark blue space filled with white stars of various sizes. In the top left is a large planet with pink and orange horizontal stripes. In the top center is a yellow comet with a purple nucleus and a long, multi-colored tail. In the top right is a large yellow star. In the bottom right is a large planet with blue and purple horizontal stripes. In the bottom left are several yellow stars of different sizes. A small pink planet with a blue ring is located near the bottom left of the text.

FINAL CHALLENGE

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GOALKEEPER

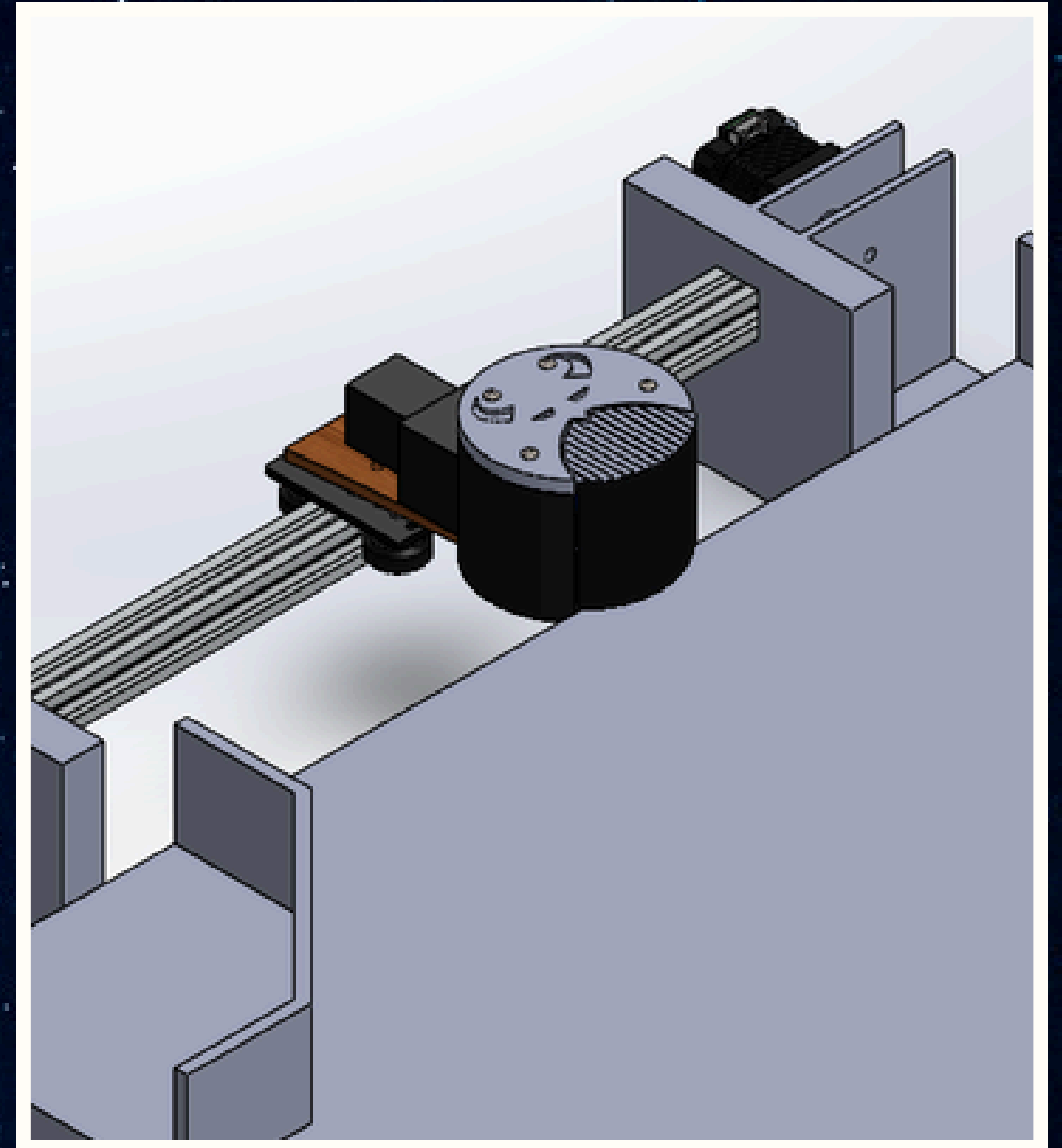
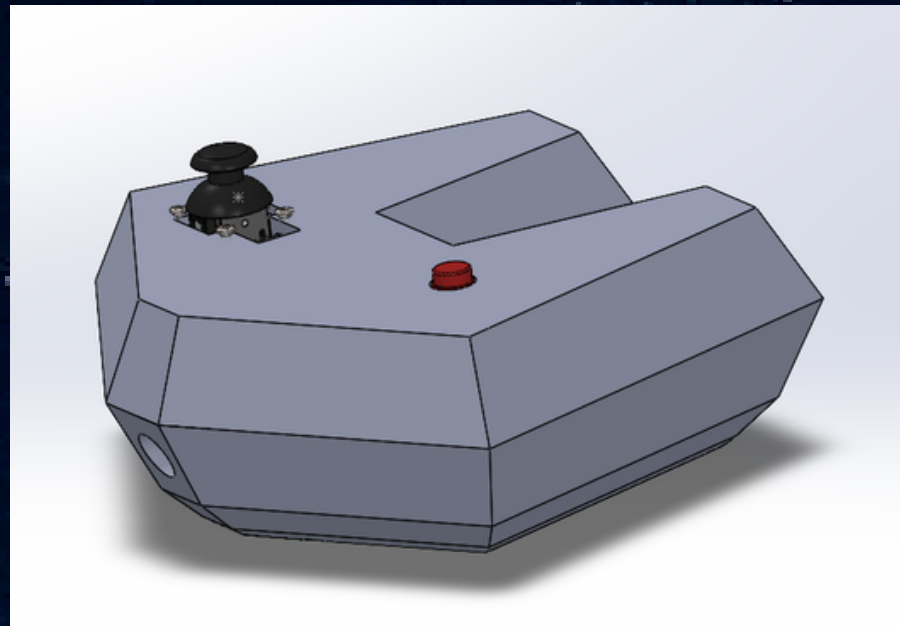
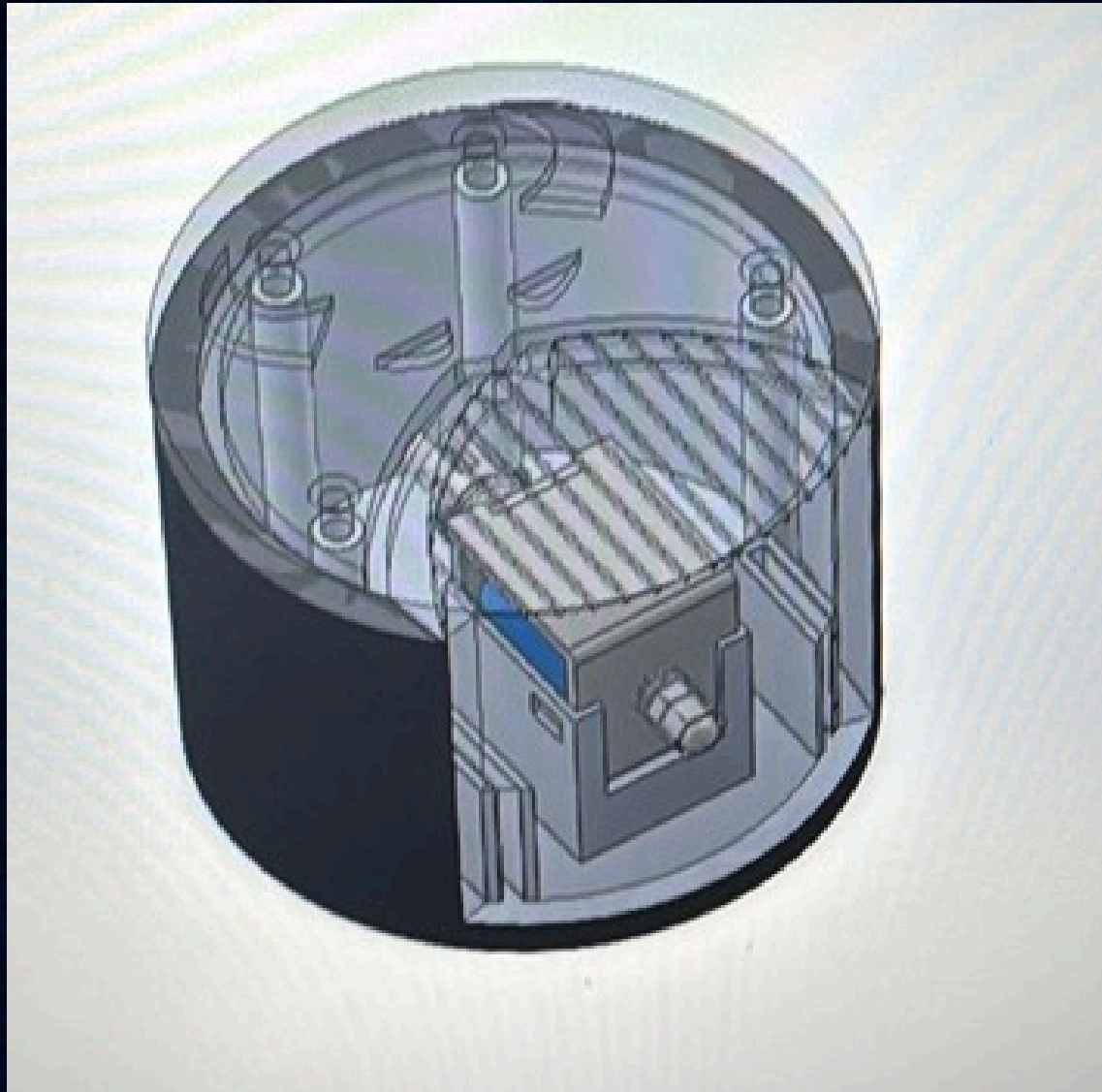


The objective of this project was to design and build an interactive robotic goalkeeper capable of moving laterally along the goal while simulating a kicking motion using a solenoid actuator.

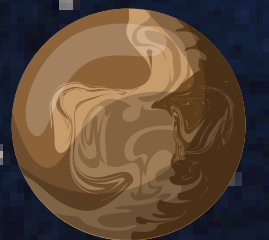
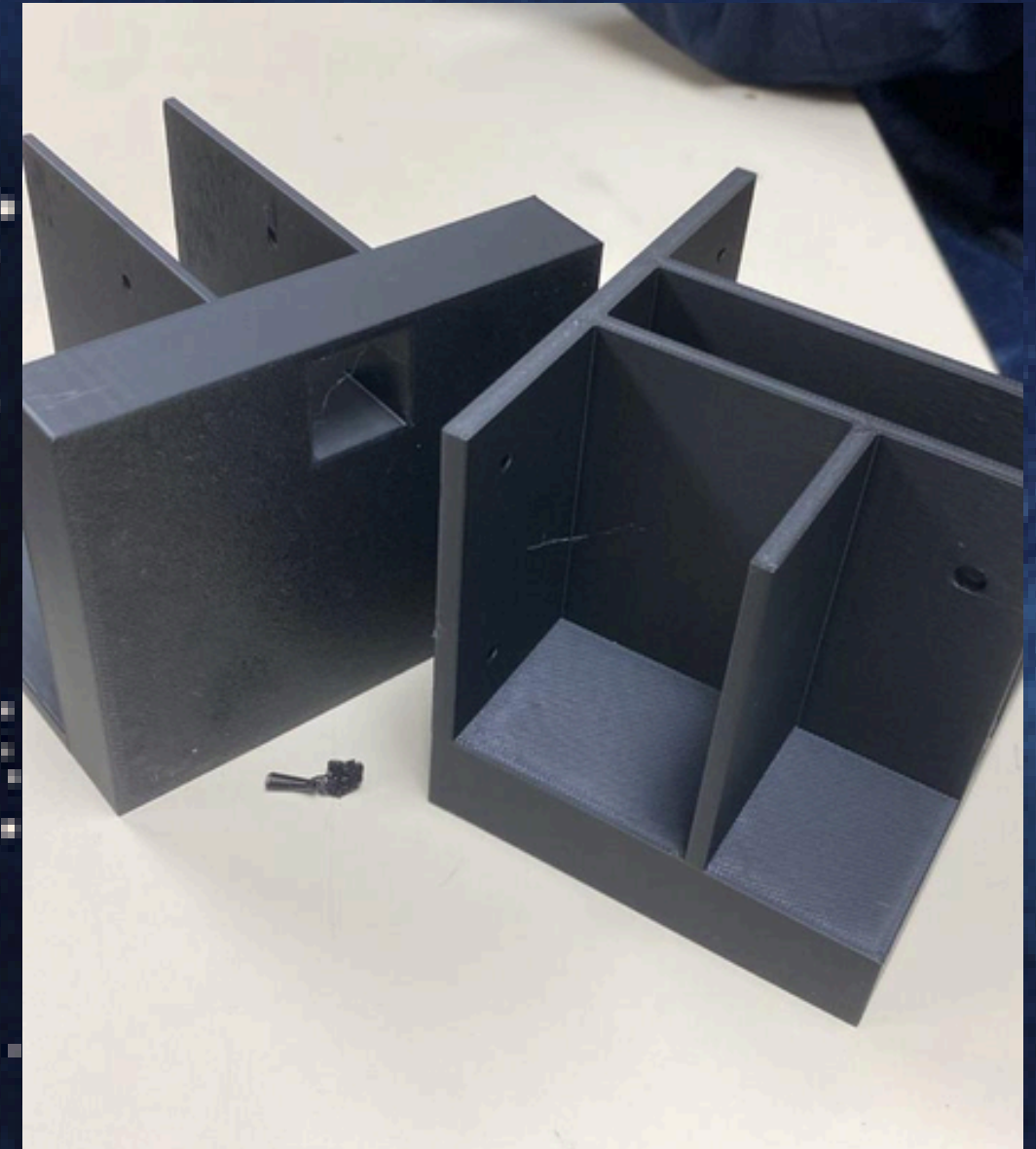
The system combines an Arduino-based control system, a stepper motor with linear motion transmission, a custom electronic circuit, and 3D-printed mechanical components.

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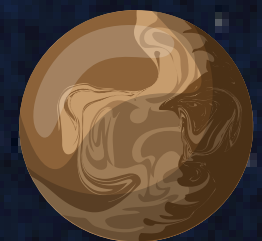
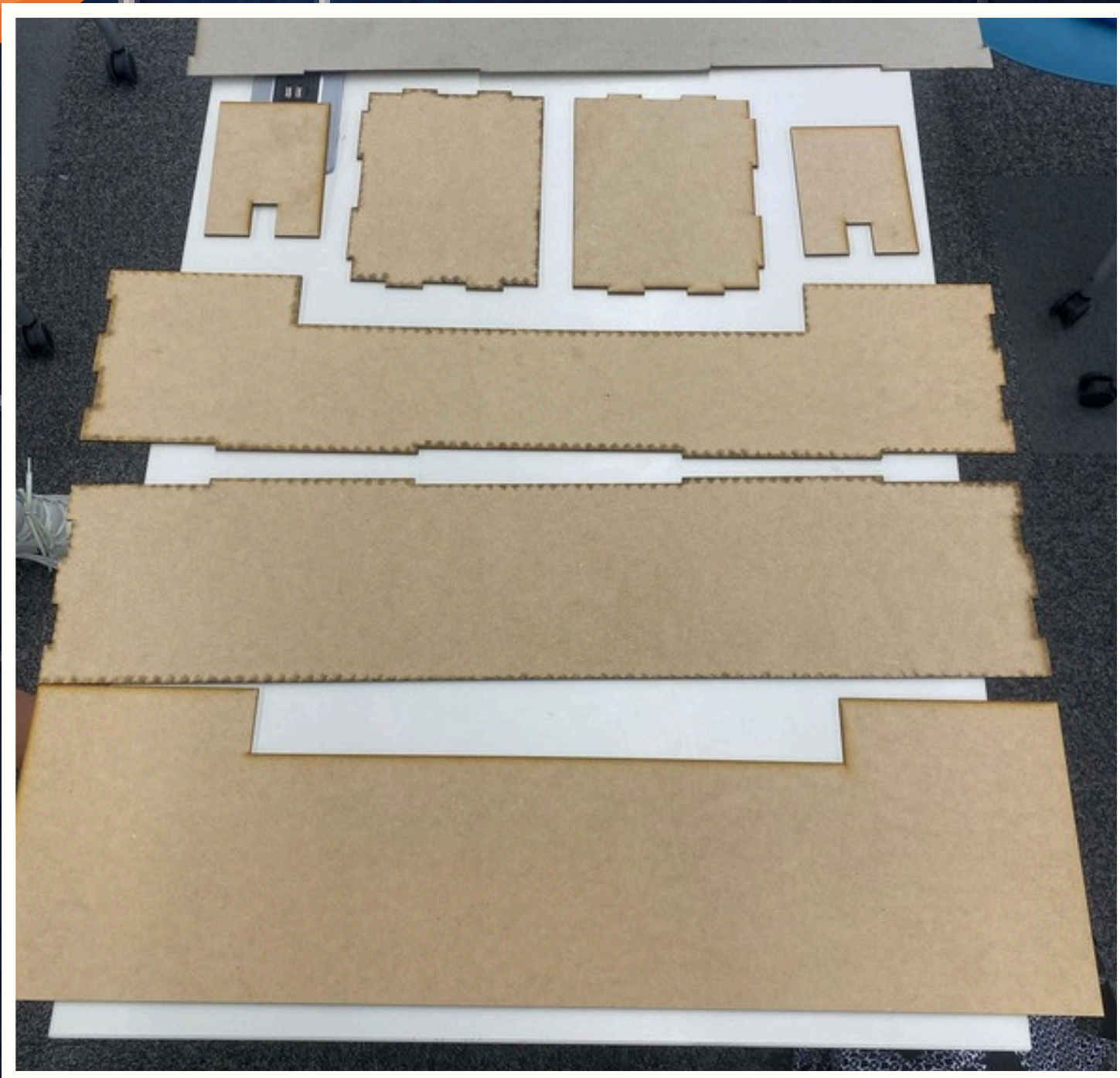
CAD



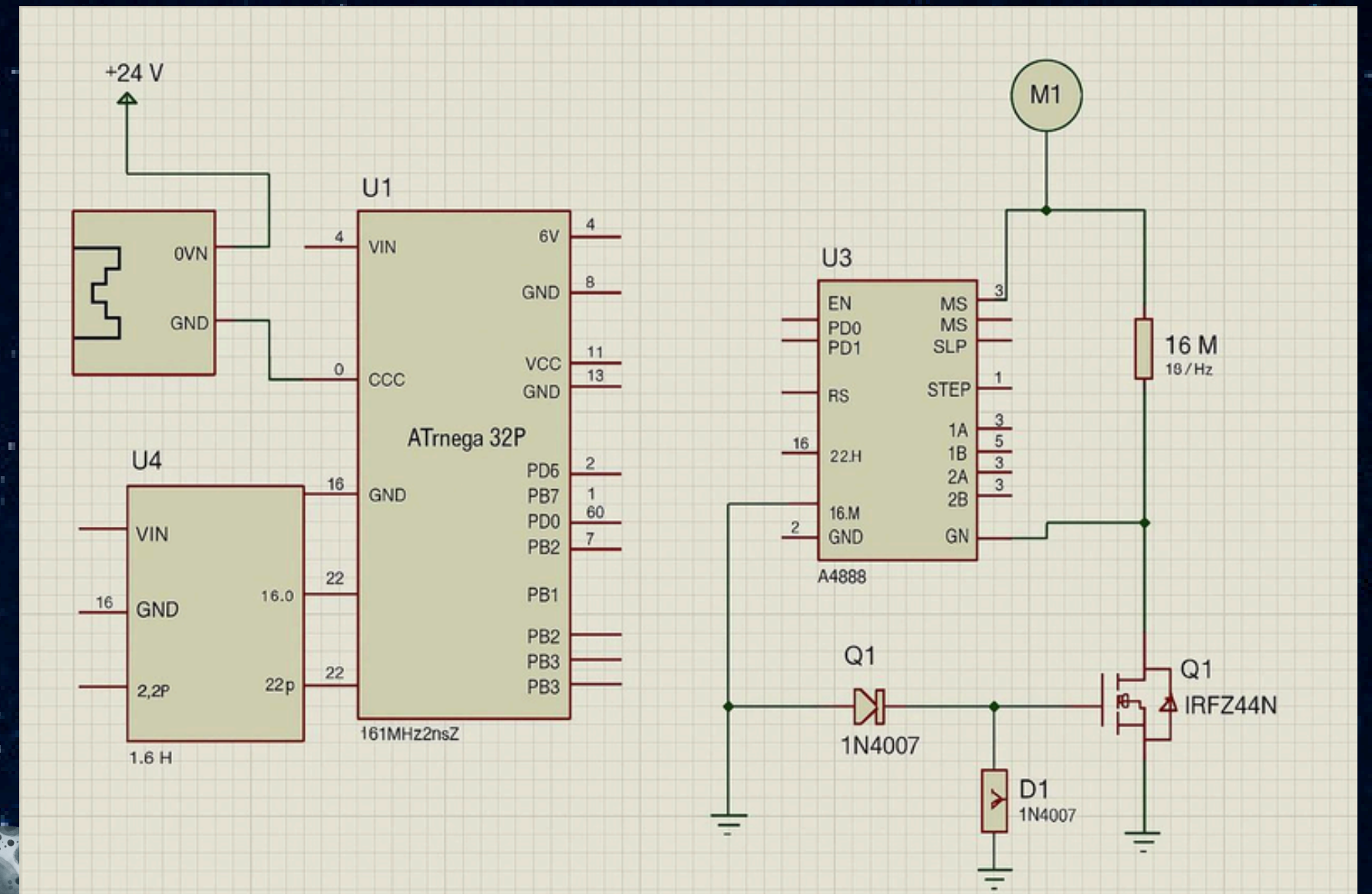
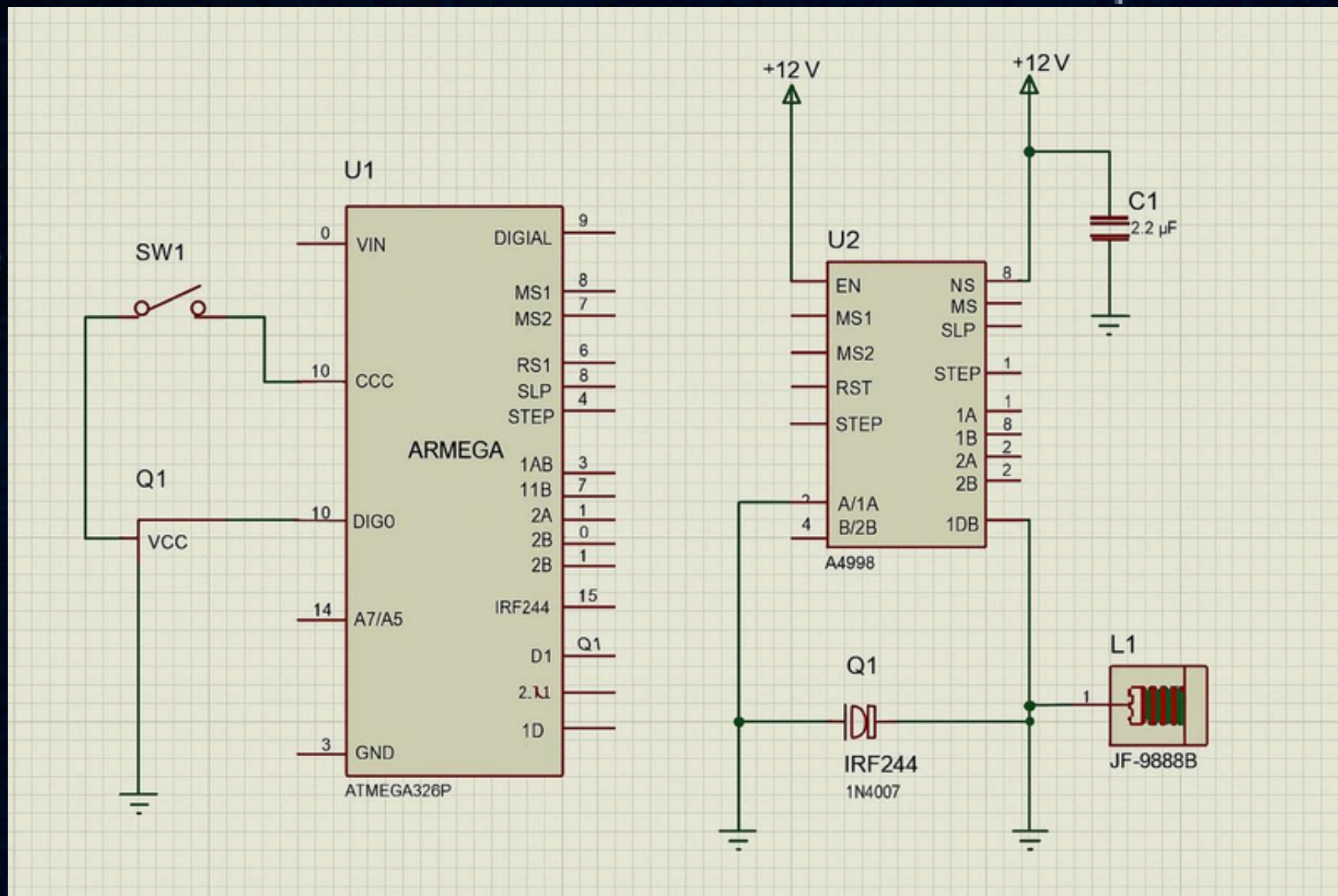
MANUFACTURE



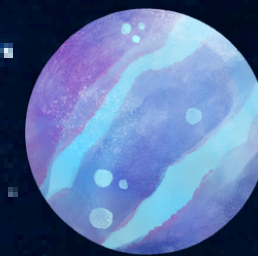
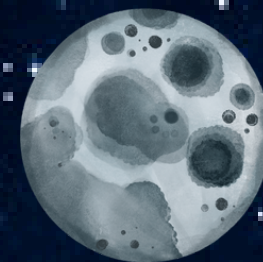
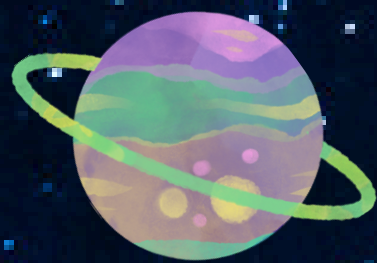
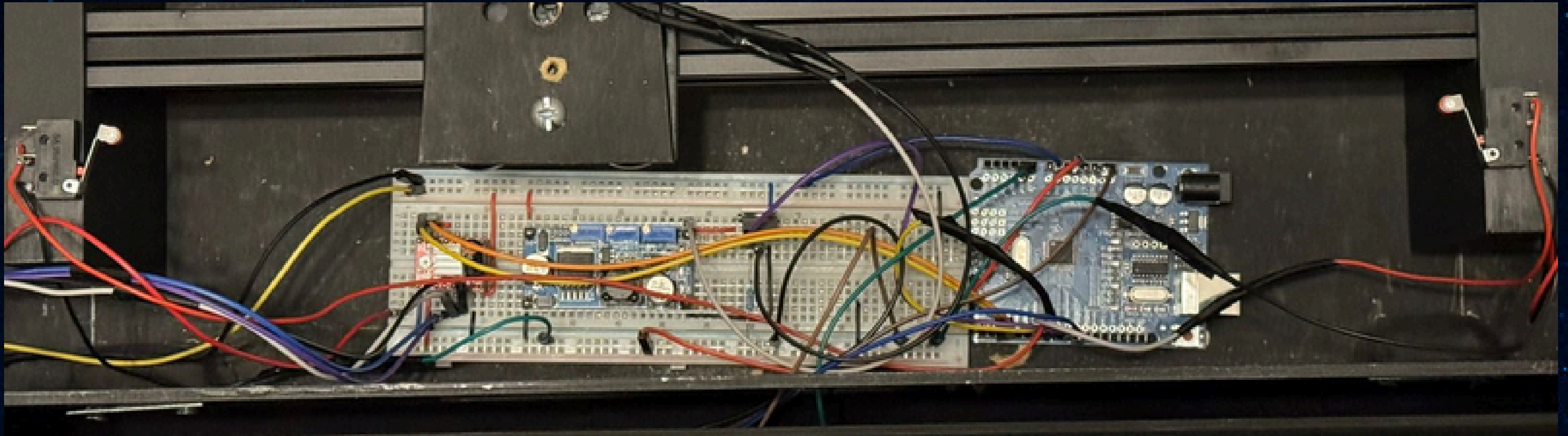
MANUFACTURE



ELECTRONICS



ELECTRONICS



CODE

GOALKEEPER_FINAL_CODE.ino

```
1  const int dirPin = 2;
2  const int stepPin = 3;
3
4  int xPin = A0;
5  int xVal;
6
7  int ButtonPin = 10;
8  int ButtonState = 0;
9
10 int MosfetPin = 9;
11 int LeftSwitchPin = 7;
12 int RightSwitchPin = 6;
13
14 int LeftSwitchState = 0;
15 int RightSwitchState = 0;
16 int dt = 50;
17
18 int stepsPerRevolution = 200;
19 int stepDelay = 2;
20
21
22 void setup() {
23   pinMode(dirPin, OUTPUT);
24   pinMode(stepPin, OUTPUT);
25   pinMode(xPin, INPUT);
26
27   pinMode(ButtonPin, INPUT_PULLUP); // Botón con pullup
28   pinMode(MosfetPin, OUTPUT); // Salida para el solenoide
29
30   pinMode(LeftSwitchPin, INPUT_PULLUP);
31   pinMode(RightSwitchPin, INPUT_PULLUP);
32
33   digitalWrite(MosfetPin, LOW); // Asegura que el solenoide no se active al inicio
```

GOALKEEPER_FINAL_CODE.ino

```
34
35   Serial.begin(9600);
36 }
37
38 void loop() {
39
40   ButtonState = digitalRead(ButtonPin);
41
42   // Control del solenoide
43   if (ButtonState == LOW) { // Botón presionado
44     digitalWrite(MosfetPin, HIGH); // Activa el solenoide
45   } else {
46     digitalWrite(MosfetPin, LOW); // Desactiva el solenoide
47   }
48
49   xVal = analogRead(xPin);
50   LeftSwitchState = digitalRead(LeftSwitchPin);
51   RightSwitchState = digitalRead(RightSwitchPin);
52
53   // Depuración
54   Serial.print("X value = ");
55   Serial.print(xVal);
56   Serial.print(" - Left Switch = ");
57   Serial.print(LeftSwitchState);
58   Serial.print(" - Right Switch = ");
59   Serial.print(RightSwitchState);
60   Serial.print(" - Button = ");
61   Serial.println(ButtonState);
62
63   // Control del motor a pasos basado en el joystick
64   if (RightSwitchState == HIGH && xVal < 450) {
65     digitalWrite(dirPin, HIGH); // Derecha
66     for (int i = 0; i < 100; i++) {
```

```
67     digitalWrite(stepPin, HIGH);
68     delayMicroseconds(1000);
69     digitalWrite(stepPin, LOW);
70     delayMicroseconds(1000);
71   }
72 }
73 } else if (LeftSwitchState == HIGH && xVal > 650) {
74   digitalWrite(dirPin, LOW); // Izquierda
75   for (int i = 0; i < 100; i++) {
76     digitalWrite(stepPin, HIGH);
77     delayMicroseconds(1000);
78     digitalWrite(stepPin, LOW);
79     delayMicroseconds(1000);
80   }
81 }
82 }
```


FINAL MODEL



THANK
YOU

